BCS theory and Mattis-Bardeen formula

...BCS SHORT REVIEW....

The Mattis-Bardeen formula (MB) describes the complex conductuvity $\sigma(\omega, T)$ of superconductors in the BCS theory framework, so it is accurate for weak-coupling superconductors materials. The originally form was derived by Mattis and Bardeen in 1958, and also assumes the clean limit $l \to \infty$ and the Pippard limit $\xi \leq \lambda_L$. Then Nam showed in 1967 that the formula is valid also in the dirty limit $l \to 0$ and the local limit $\xi \gg \lambda_L$.
The MB equation are given by the two integrals

$$\begin{aligned}\frac{\sigma_1(\omega,T)}{\sigma_n} = \frac{2}{\hbar\omega}\int_{\Delta}^{\infty} \frac{[f(E)-f(E+\hbar\omega)]\left[E^2+\Delta^2+\hbar\omega E\right]}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE \\ + \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega}^{-\Delta} \frac{[1-2f(E+\hbar\omega)]\left[E^2+\Delta^2+\hbar\omega E\right]}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE,\end{aligned} \tag{1}$$

for the real part and by

$$\frac{\sigma_2(\omega,T)}{\sigma_n} = \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega,\,-\Delta}^{\Delta} \frac{[1-2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{\Delta^2-E^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE, \tag{2}$$

for the imagimary part, where the σ_n is the normal state conductivity and $f(E) = \frac{1}{1+exp(\frac{E}{TK_B})}$ is the Fermi-Dirac distribution. The lower limit in the integral of Eq. 2 is $-\Delta$ if $\hbar\omega > 2\Delta$.

...SOME COMMENTS ABOUT THE EXPRESSIONS ...

Numerical Integration of the Mattis-Bardeen formula

Clearly Eqs.2 and 1 are not analytically integrable, so we need to use numerical methods to evaluate them. In this work i used the standard integrating methods in MATLAB, being careful with the singularities in the limits of integration. For the sake of simplicity

of notation let's define the following integrals from the MB equations:

$$I_1 = \frac{2}{\hbar\omega}\int_{\Delta}^{\infty} \frac{[f(E) - f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE, \tag{3}$$

$$I_2 = \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega}^{-\Delta} \frac{[1 - 2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE \quad \text{and} \tag{4}$$

$$I_3 = \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega,-\Delta}^{\Delta} \frac{[1 - 2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{\Delta^2-E^2}\sqrt{\Delta^2-(E+\hbar\omega)^2}}\,dE. \tag{5}$$

The problems are in the square root terms in the denominators, which lead to singularities. To overcome this issue, i used a change of variable in the integrals that cancels out the singular terms in the denominator.

For the integral I_1 we have a problem in the lower limit $E = \Delta$, but using

$$E = \Delta\ cosh(u) \quad \leftrightarrow \quad u = cosh^{-1}(\frac{E}{\Delta}),$$

so that

$$du = \frac{dE}{\sqrt{E^2-\Delta^2}} \quad \Rightarrow \quad dE = \sqrt{E^2-\Delta^2}du$$

and the integral becames

$$I_1 = \frac{2}{\hbar\omega}\int_{0}^{\infty} \frac{[f(\Delta cosh(u)) - f(\Delta\ cosh(u)+\hbar\omega)]\left[\Delta^2(cosh^2(u)+1)+\hbar\omega\Delta\ cosh(u)\right]}{\sqrt{(\Delta\ cosh(u)+\hbar\omega)^2-\Delta^2}}\,du. \tag{6}$$

Now we can integrate numerically with the standard methods in MATLAB sush as .

The integrals I_2 and I_3 have singularities both in the lower and upper limits, we can write them as

$$\frac{1}{\hbar\omega}\int_{a}^{b} \frac{[1 - 2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{b^2-E^2}\sqrt{(E^2-a^2}}\,dE. \tag{7}$$

Then with the change of variables

$$E = \frac{b-a}{2}\ sin(t) + \frac{b+a}{2} \quad \leftrightarrow \quad t = sin^{-1}\left(\frac{2E-(b+a)}{b-a}\right),$$

that leads to

$$dt = \frac{dE}{\sqrt{b-E}\sqrt{E-a}} \Rightarrow dE = \sqrt{(b-E)(E-a)}dt,$$

we can rewrite Eq. 7 as

$$\frac{1}{\hbar\omega}\int_{-\pi/2}^{\pi/2} \frac{\left[1 - 2f\left(\frac{b-a}{2}\ sin(t)+\frac{b+a}{2}+\hbar\omega\right)\right]\left[\left(\frac{b-a}{2}\ sin(t)+\frac{b+a}{2}\right)^2+\Delta^2+\hbar\omega\left(\frac{b-a}{2}\ sin(t)+\frac{b+a}{2}\right)\right]}{\sqrt{(b+E)(E+a)}}\,dt. \tag{8}$$

And the singularities have been removed.

Results with BCS theory

Now that we can solve Eqs. 1 and 2 let's see the results for some different materials and various temperature. In particular i have tested Nb, Pb and Al.

For different materials we need the experimental values of thecritical temperature T_c, under which they become superconductors, and the zero temperature value of the enrgy gap Δ_0. Actually for Δ_0 we extract the value from the BCS relation

$$\frac{2\Delta_0}{K_B T_c} = \alpha, \tag{9}$$

where α is ~ 3.53 for weak-coupling materials, that is the limit where BCS theory and MB formulas holds, and is greater for strong-coupling materials such as Nb and Pb. In Tab. 1 below the T_c and α values are showed.

Since $\Delta = \Delta(T)$ is a function of the temperature and is zero at $T \geq T_c$ i have extrapolate it in the BCS approximation as

$$\Delta = \Delta_0 \ \tanh\left(e^{\gamma}\sqrt{\frac{8}{7\xi(3)}}\ \sqrt{\frac{T_c}{T}-1}\right) \simeq \Delta_0 \ \tanh\left(1.74\ \sqrt{\frac{T_c}{T}-1}\right). \tag{10}$$

In the following we see some test for Nb, Pb and Al at different temperatures. In Fig. 1 i compared the real and imaginary conductivity for Nb, Pb and Al at $T = 0.8T_c$. In Figs.4, 2 and 3 we see conductivity for different temperatures in the various materials. The last Fig. 5 i have plotted the conductivity of Niobium with the energy scale normalized by $2\Delta_0$ to check if the real part of σ/σ_n correctly drops down at $\hbar\omega = 2\Delta_0$ when $T = 0$, and if increasing T the gap moves to lower energies/frequencies.

Table 1: Superconducting materials experimental parameters

Material	T_c(K)	α
Nb	9.3	3.65
Pb	7.2	3.95
Al	1.2	3.53

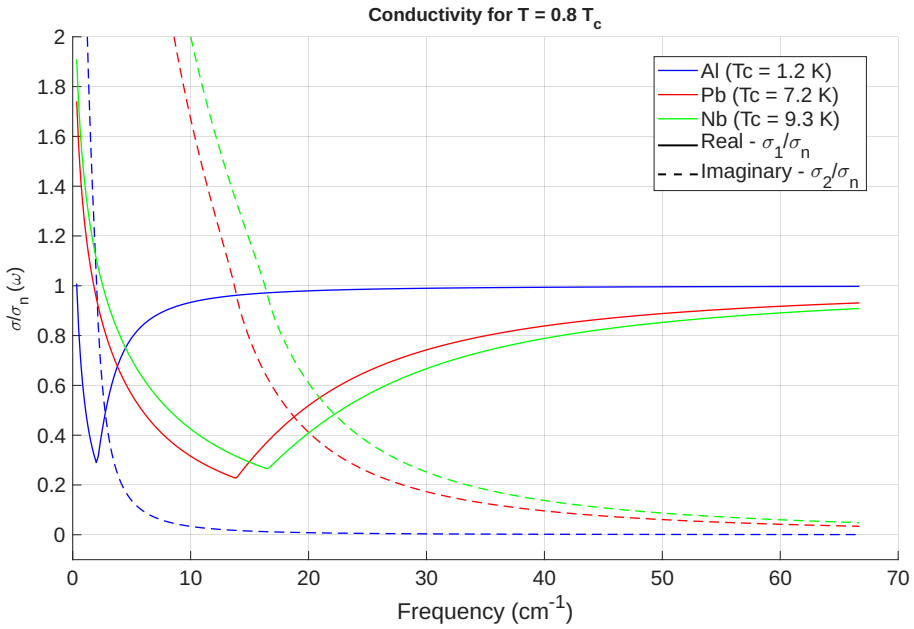


Figure 1: Complex conductivity for different materials.

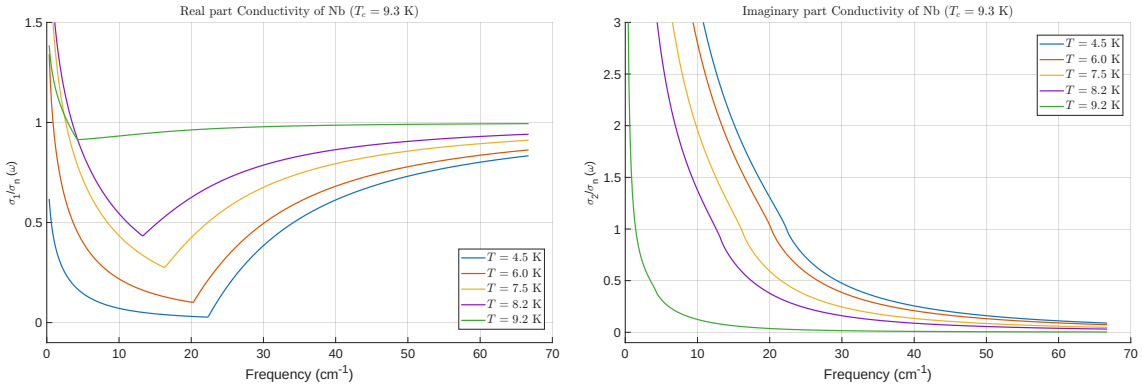


Figure 2: Real (sx) and imaginary (dx) conductivity for Nb.

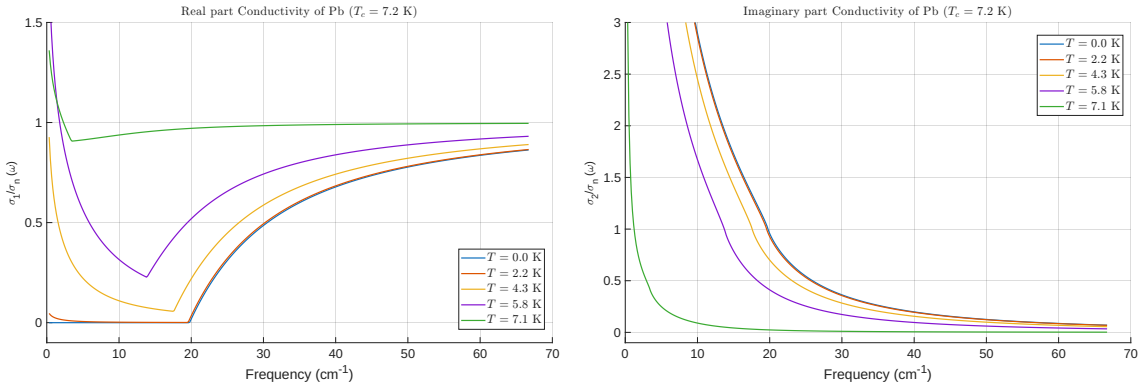


Figure 3: Real (sx) and imaginary (dx) conductivity for Pb.

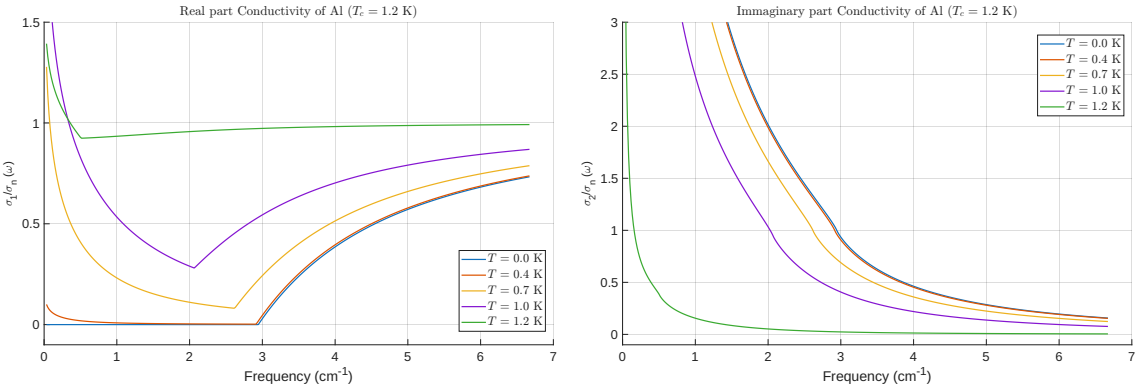


Figure 4: Real (sx) and imaginary (dx) conductivity for Al.

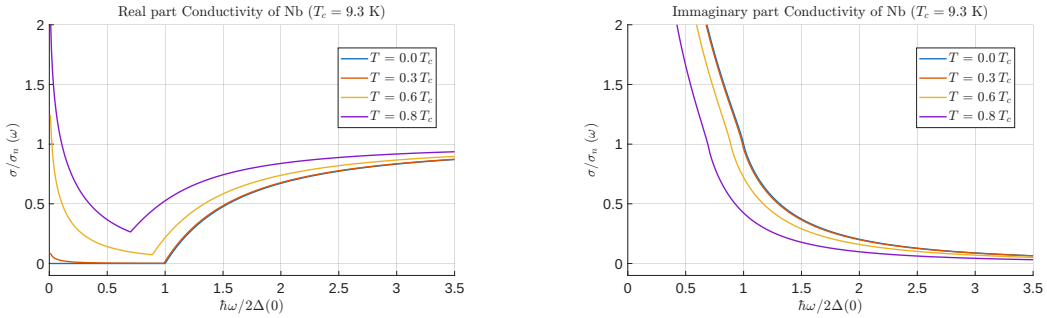


Figure 5: Real (sx) and imaginary (dx) conductivity for Nb in normalized units, to check the gap corresponds at $T = 0$.